

Data Analyst

Syllabus



TECHTONIC-LAB

Tools & Major Topics Covered

- A. Introduction to Data Analysis
- B. SQL
- C. Power Bi/Tableau (*Choose one – Power Bi recommended*)
- D. Python
- E. Statistics
- F. Advance Excel
- G. Introduction to cloud services (AWS/Azure) - *Bonus*
- H. End-End Project

A. Introduction to Data Analysis

1. What is Data & Data Analytics?
2. Global Scope of Data Analysis
3. Data Analytics in Different Domains
4. Understanding the vision of stakeholders and asking effective questions
5. Road Map to be a Data Analyst
6. Role Of Data Analyst?
7. Data Analyst vs Business Analyst
8. What is BI and Its Importance?
9. BI Tools and Techniques
10. Data Analysis Methods
11. Types of Data Analysis
12. Data Analysis Process

B. SQL

- **Introduction to SQL**
 - 1) Data Base & Types
 - 2) DBMS vs RDBMS
 - 3) Installation
 - 4) MySql User Interface
- **SQL Commands**
 - 1) DDL
 - 2) DML
 - 3) DQL
 - 4) TCL
 - 5) DCL
- **SQL Data Types**
 - 1) Numerical

- 2) Character
- 3) Date
- 4) DateTime
- 5) Bool

- **Data Loading**

- 1) Table Import Wizard
- 2) Using Query

- **Operators in SQL**

- 1) Arithmetic operators
- 2) Comparison operators
- 3) Logical operators
- 4) Operators used to negate conditions

- **Constraints**

- 1) Primary Key
- 2) Foreign Key
- 3) Unique
- 4) DEFAULT
- 5) Not Null
- 6) Check

- **Advanced constraints**

- 1) Artificial Key
- 2) Composite Key
- 3) Super Key
- 4) Index

- **Set Operators**

- 1) Intersect
- 2) Except
- 3) Union
- 4) Union all

- **Joins**

- 1) Self Join
- 2) Inner Join
- 3) Left Join
- 4) Right Join
- 5) Full Outer join
- 6) Cross Join
- 7) Natural Join

- **Subqueries**
 - 1) Using From and Where Clause
 - 2) Correlated Subqueries
 - 3) Filter Query Results using Query from Different Table
 - 4) Order of execution
- **Case Statement and Temporary Tables**
 - 1) Case Statement
 - 2) Temporary Table
 - 3) CTE
 - 4) Recursive CTE
- **Window Functions**
 - 1) Partition
 - 2) Rank and Dense Rank
 - 3) Row Number
 - 4) Lead and Lag Functions
 - 5) Min, Max, Avg, Count
 - 6) Unbounded Preceding, Preceding, Following
- **Stored Procedures**
 - 1) Return Type - Non-Parametric
 - 2) Return Type - Parametric
 - 3) Non-Return Type - Non-Parametric
 - 4) Non-Return Type – Parametric
- **Functions**
 - 1) System Functions
 - 2) User-Defined Functions
 - 3) Predefined Functions
 - 4) Arithmetic Functions
 - 5) Logical Functions and Error Functions
 - 6) Date Functions
 - 7) Mathematical Functions
 - 8) Text Functions
 - 9) Statistical Functions
 - 10) Database Functions and Array
 - 11) Introduction to Functions
 - 12) Built-in Functions
 - 13) User-Defined Functions
 - 14) Anonymous Functions (Lambda Functions)
 - 15) Recursive Functions
- **Views and Cursor**

C. Power Bi

- **Introduction to Power BI**
 - 1) What is Power BI
 - 2) Power BI Products
 - 3) Components of Power BI
 - 4) Why Power BI - PowerBI Vs Other BI Tools
 - 5) Building Blocks of Power BI
 - 6) Power BI Process
 - 7) Power BI Desktop Installation
 - 8) User Interface
- **Power Query Editor**
 - 1) Connecting to a local data Sources
 - 2) Power Query User Interface
 - 3) data types
 - 4) Home Tab
 - 5) Transform Tab
 - 6) Add Column
 - 7) Other Transformations
- **Visualizations**
 - 1) Bar/Column charts
 - 2) Pie & Donut charts
 - 3) Waterfall chart
 - 4) Tree Map
 - 5) Table & Matrix visualization
 - 6) Ribbon chart
 - 7) Funnel
 - 8) Line
 - 9) Slicers
 - 10) KPI
 - 11) Area
 - 12) Cards
 - 13) Scatter
 - 14) Gauge Chart
 - 15) Maps
 - 16) Custom Visuals (Sankey, Sunburst, Playaxis, Wordcloud, Butterfly)
- **Data modelling**
 - 1) Data models, Importance
 - 2) Schema
 - 3) Creating and Managing relationships
 - 4) Optimizing Data Models

- **DAX: basics**
 - 1) DAX and its Importance
 - 2) Row & Filter context
 - 3) Measures & Calculated columns

- **DAX: Aggregation functions**
 - 1) SUM function
 - 2) SUMX function
 - 3) MIN function
 - 4) MINX function
 - 5) MAX function
 - 6) MAXX function
 - 7) AVERAGE function
 - 8) AVERAGEX function
 - 9) COUNT function
 - 10) DISTINCTCOUNT function

- **DAX: Filter functions**
 - 1) ALLSELECT function
 - 2) FILTER function
 - 3) CALCULATE function

- **DAX: Relationship functions**
 - 1) CROSSFILTER function
 - 2) RELATED function

- **DAX: Date & Time functions**
 - 1) CALENDAR function
 - 2) DATE function
 - 3) DATEDIFF function
 - 4) MONTH function
 - 5) WEEKDAY function
 - 6) YEAR function

- **DAX: Time intelligence functions**
 - 1) DATEADD function
 - 2) DATESBETWEEN function
 - 3) TOTALMTD function
 - 4) TOTALQTD function
 - 5) TOTALYTD function

- **DAX: Text functions**

- 1) CONCATENATE function
- 2) CONCATENATEX function
- 3) FORMAT function
- 4) LEFT function
- 5) RIGHT function
- 6) TRIM function
- 7) FIND function
- 8) SEARCH function

- **DAX: Logical functions**

- 1) AND function
- 2) OR function
- 3) NOT function
- 4) FALSE function
- 5) TRUE function

- **Connectivity**

- 1) Introduction to using Excel data in Power BI
- 2) Exploring live connections to data with Power BI
- 3) Connecting directly to Database
- 4) Import Power View and Power Pivot to Power BI
- 5) Power BI Publisher for Excel
- 6) Content packs
- 7) Introducing Power BI Mobile

- **Creating reports**

- 1) What is a report ?
- 2) Filters; Visual level filter
- 3) Filters; Page level filter
- 4) Filters; Report level filter
- 5) Filters; Drill through filter
- 6) Include, Exclude Filter
- 7) Formating a report

- **Security**

- Row level security

- **Publishing, sharing & scheduling a refresh**

- 1) Creating an account on PBI service
- 2) Introduction to PBI Service
- 3) Publishing a report to PBI Service
- 4) Creating a dashboard
- 5) Creating a Gateway
- 6) Scheduling a refresh

D. Tableau

- **Tableau introduction**
 - 1) How tableau can be implemented in Data Industry
 - 2) Why Tableau?
 - 3) Live vs Extract
 - 4) Data source overview
- **Tableau download and Installation**
 - 1) Tableau Desktop Download for Windows
 - 2) Tableau Desktop download for MAC OS
- **How To Connect Tableau With Different Data Sources**
 - 1) Connecting to Excel and CSV files
 - 2) Connecting to Microsoft SQL Server
 - 3) Connecting to Microsoft Access
 - 4) Connecting to PDF file
 - 5) Connecting to Google Drive
 - 6) Connecting to MongoDB through ODBC/ JDBC
 - 7) Connecting to Web Database Connector (WDC)
- **Tableau Architecture & Products**
 - 1) Architecture
 - 2) Tableau Server
 - 3) Tableau Cloud (Online)
 - 4) Tableau Public Server
 - 5) Tableau desktop
 - 6) Tableau public desktop
 - 7) Tableau reader
 - 8) Tableau mobile
 - 9) Tableau prep builder
- **In-built Charts**
 - 1) Area charts
 - 2) Bar chart
 - 3) Bubble chart
 - 4) Box & whisker plot
 - 5) Dual combination chart
 - 6) Bullet graph
 - 7) Heat map
 - 8) Line chart
 - 9) Pie chart
 - 10) Scatter plot

- 11) Stacked bar chart
- 12) Treemap
- 13) Geographic & filled map

- **Advanced charts**

- 1) bump chart
- 2) Funnel chart
- 3) Gantt chart
- 4) Sunburst
- 5) Lollipop
- 6) Word cloud
- 7) Pareto chart
- 8) Donut
- 9) Dumbbell
- 10) waterfall chart

- **Reference types**

- 1) Reference lines
- 2) Reference bands
- 3) Distribution bands

- **Filters**

- 1) Types
- 2) Extract, Data sources, Dimension & measures
- 3) Cascading Filter
- 4) Context Filter
- 5) Order of operation of filters

- **Tableau calculations using functions**

- 1) Number function
- 2) String function
- 3) Date function
- 4) Type conversion
- 5) Logical functions
- 6) Aggregate functions
- 7) User function
- 8) Table calculation function

E. Python

- Introduction
 - 1) Installation guide of Jupyter Notebook and use of Google Colab
 - 2) Working with Python notebooks
 - 3) Data Types/structures :-
 - I. Lists
 - II. Tuple
 - III. Dictionary
 - IV. Set
 - 4) Python Loops & Functions:-
 - I) Branching (If Else Elif)
 - II) Range
 - III) Loops
 - IV) Indexing
 - V) Break and Continue
- Exception Handling
- Class & Object
 - 1) Numpy:- Introduction to Numpy
 - 2) Data types
 - 3) Arange and space
 - 4) Matrix creation
 - 5) Random number generation
 - 6) Reshaping
 - 7) Indexing and slicing
 - 8) Subsetting
 - 9) Universal Functions
 - 10) Broadcasting
 - 11) Array Math
- Pandas
 - 1) Introduction to Pandas
 - 2) Pandas Operations
 - 3) Read and write from CSV and TSV files
 - 4) Read and write operations from HTML
 - 5) Read and write operations from web API
 - 6) Creating and working with Data frames
 - 7) Date time manipulation
 - 8) Data Cleaning
 - 9) Data manipulation
 - 10) Series manipulation
 - 11) Data frame manipulation

- **Matplotlib**

- 1) Introduction to Matplotlib & Seaborn
- 2) SubPlot, ScatterPlot, Area Plot,
- 3) Bar graph, Stem Plot, Step_plot, fill_Between Plot
- 4) imshow(), pcolor mesh(), Flat shading, contour plot, contourf plot
- 5) Barps Plot, Quiver, Stream Plot

- **Statistical Plot**

- 1) Histogram, Box Plot, Error Bar Plot, Violin Plot
- 2) Event Plot, Hist 2D, Hex Bin Plot, Pie

F. Statistics

- **Statistics**

- 1) Types of Statistics
- 2) Descriptive
- 3) Inferential
- 4) Population and Sample
- 5) Parameter and Statistics\n(Mean, Median, Mode, Std, Variance)
- 6) Uses of variable
- 7) Dependent
- 8) Independent variable
- 9) Types of Variable
- 10) Continuous
- 11) Categorical variable
- 12) Distribution Types and Skewness
- 13) Hypothesis testing
- 14) Type 1 Error
- 15) Type 2 Error
- 16) Ttest (one sample and sample)
- 17) ANOVA & CHi-SQUARE

- **Covariance and Correlation**

- 1) Views
- 2) Cursor

- **Triggers**

- 1) For Insert
- 2) For Update
- 3) For Delete

- **Data Modelling**

- 1) Introduction to ER diagram
- 2) Mapping Constraints

- 3) Cardinality
- 4) Relational Model Concept

- **Normalization**

- 1) 1NF
- 2) 2NF
- 3) 3NF
- 4) BCNF

G. Excel

- **Excel Interface / Basic Parts in an MS-Excel window**

- 1) Launching Excel
- 2) Quick access toolbar, Title Bar, Ribbon
- 3) Menu Bar, Formula Bar, Status Bar
- 4) Scroll Bar, Column Bar, Row Bar, Leaf Bar
- 5) Name box, Workspace, Dialog box launcher, Zoom controls, View buttons
- 6) Tabs - File, Home, Insert, Page Layout, Formulas, Data, Review, View, Help

- **Row, Column, Cell, Active Cell**

- 1) Row number, Column Name
- 2) Inserting and Deleting Columns and Rows
- 3) Range of cells
- 4) Inserting & Deleting Cells
- 5) Inserting Multiple Columns & Rows
- 6) Modifying Cell Width and Height
- 7) Hiding and Unhiding Rows and Columns
- 8) Freeze panes

- **Look Up**

- 1) Lookup
- 2) Vlook up
- 3) Hlookup
- 4) Xlookup
- 5) Index
- 6) Index and Match
- 7) Offset

- **Data Validation**

- **Visualization**

- **Summarization**

- 1) Pivot Tables
- 2) Slicers
- 3) Dashboard creation
- 4) Formatting Dashboard

- **Macros and VBA**
 - 1) What is Macro?
 - 2) Recording a Macro
 - 3) Adding a Developer Tab
 - 4) Macros by button
 - 5) Editing Macros
 - 6) VBA - Subs, Ranges and sheets
 - 7) If statements and conditions
 - 8) Loops
 - 9) Message Boxes and Input Boxes